

Penalty view



SafeOT-Dual

One price rule, two views

Constraint view



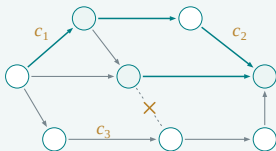
Encode jointly



Shared state space

One policy update

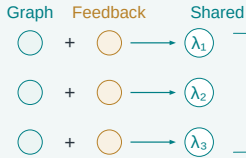
Price on one graph



Budgets + flow balance
× excluded graph edge

Reward + cost advantages

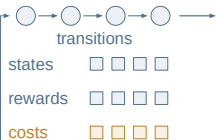
Combine prices



π PPO / TRPO

Observe

Policy rollout



Episode costs

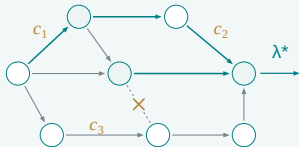


GAE advantages

Schematic

Construct shared prices

Joint graph



Solve jointly

Balance · per-step budgets
Newton / bisection

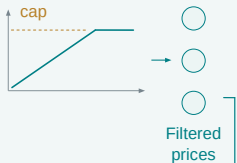
Episode-cost feedback



cost - budget

Accumulate

Clip + smooth



Filtered prices

$\times$ hard graph support

β



Update policy

Reward advantage

A_r

Cost advantages

A_c

Shared prices

$\times$

$+$

$-$

Normalize

PPO / TRPO

π

next rollout